

Title

Antenatal Aerobic Exercise and Maternal-Neonatal Outcomes: Exploratory Secondary Findings

Abstract

Understanding the broader health impacts of structured prenatal exercise extends beyond mood outcomes to encompass maternal metabolism, infant development, and delivery-related complications. We conducted secondary analyses of maternal weight gain, gestational glucose tolerance, delivery complications, and neonatal birthweight and health indicators in our trial participants. Pregnant women assigned to supervised aerobic activity demonstrated beneficial trends in gestational weight accumulation and metabolic parameters, though these differences did not reach statistical significance in this secondary analysis. Delivery method and perinatal complications were equally distributed across groups. Neonatal outcome measures including birthweight percentiles, Apgar scores, and short-term morbidity were comparable and reassuring across both arms. These secondary findings confirm that structured aerobic exercise during pregnancy is a safe intervention without adverse neonatal consequences, and suggest potential ancillary maternal metabolic benefits that warrant investigation in larger-scale studies.